# Real-Time Multimodal AI Framework for Student Engagement Detection and Adaptive Assessment Generation Using CNN, NLP, and Computer Vision


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***Abstract*—This paper presents an AI-driven educational platform that combines real-time attention monitoring with adaptive assessment to create a personalized and engaging learning experience. The system tracks student engagement using webcam-based facial analysis, employing Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), head pose estimation, and a custom CNN to detect drowsiness, yawning, absence, and inattention. Built with Flask, TensorFlow/Keras, and MediaPipe, it provides real-time feedback, session management, and a web-based interactive interface. In the evaluation stage, uploaded materials are processed using OCR and content analysis to extract key concepts and generate adaptive quizzes and concept network visualizations. To sustain motivation and focus, the platform incorporates gamification, study notes, focus music, and live chat agents for real-time guidance, while dashboards allow teachers and parents to monitor progress and provide personalized feedback. This scalable framework bridges behavioral monitoring, intelligent content processing, and adaptive evaluation to foster effective, collaborative, and enjoyable learning experiences.**



***Index Terms*—eal-time engagement monitoring, Affective computing, Document intelligence, Concept clustering, Adaptive assessment, Deep learning, Computer vision, Natural language processing, Adaptive Assessment, Gamification, Personalized Learning.**


## I. Introduction

Artificial intelligence is transforming education by enabling personalized, real-time adaptation in both classrooms and self-paced learning. Traditional approaches—manual observation or delayed analytics—struggle to provide timely, objective insights into student engagement and understanding. Multimodal learning analytics (MMLA) combines behavioral monitoring, content analysis, and adaptive assessment, yet most systems treat these separately, lacking integration between real-time engagement detection and content-aware evaluation. Current solutions have significant shortcomings. Engagement detection systems often use controlled datasets with binary labels, missing the nuanced spectrum of attention and underperforming in diverse real-world settings. They typically ignore content-driven personalization, providing generic metrics disconnected from actual learning material. Assessment platforms rely on static quizzes with fixed difficulty, failing to adapt to individual progress, context, or behavioral patterns. Most critically, they lack real-time feedback loops correlating attention with learning outcomes, leaving educators and students without actionable insights. This research introduces a unified AI platform integrating real-time attention monitoring, intelligent content analysis, and adaptive assessment. The attention monitoring uses webcam-based computer vision—Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), head pose estimation, and a custom CNN—to detect drowsiness, yawning, absence, and distraction in real time. Built with Flask, TensorFlow/Keras, and MediaPipe, it's non-intrusive and scalable. Simultaneously, the adaptive evaluation engine processes uploaded materials through OCR and semantic analysis to extract key concepts and generate personalized quizzes and concept networks matched to each learner's progress. By aligning monitoring, assessment, and feedback with actual study content, the platform ensures contextually relevant engagement detection and evaluation.

Beyond analytics, the platform includes motivation-sustaining features: gamification, focus music, structured note-taking, and live chat support. Dashboards for teachers and parents enable continuous progress tracking, connecting individual learning with broader educational oversight. By unifying monitoring, personalization, and adaptive assessment, this research enhances comprehension, supports data-driven teaching decisions, and creates engaging, learner-centered environments across educational contexts.

## II. Literature Review

Student engagement detection has become essential in online and hybrid education, driving the creation of automated, multimodal systems that assess attention and affect. This review summarizes advances in combining computer vision, facial expression analysis, and other sensor-based methods to enable intelligent, real-time engagement monitoring in digital learning environments.

TABLE I: COMPARATIVE ANALYSIS OF RESEARCH PAPERS

| **Title of Research Paper** | **Authors** | **Methodology** | **Merits** | **Engagement Analysis** | **Limitations** | **Environment** |
|---|---|---|---|---|---|---|
| Features of Students Paying and Not Paying Attention in Online Classes | Hafez et al. | Behavioral cues (eye, head, posture, facial expression, gaze) | Differentiates subtle attentive/inattentive states; non-intrusive | Gaze, head, posture (binary attention) | Limited emotion/multimodal; qualitative focus | Online classroom |
| An Adaptive System for Predicting Student Attentiveness in Online Classrooms | Gupta et al. | Adaptive Eye Aspect Ratio (AEAR), SVM/DT/RF classifiers | Personalized, high accuracy (> 92%), real-time | Eye blink/AEAR-based attentive/inattentive | Eye-focused only; small sample | Online, live video |
| Learner Attentiveness and Engagement Analysis Using Computer Vision (DAISEE-based) | Gogawale et al., DAISEE Consortium | Hybrid CNN/EfficientNet, multi-affective (DAISEE dataset) | Multi-affect classification; Attentiveness Index; dashboard | Multi-class (engage, confuse, boredom, etc.) | Not linked to outcome; requires face visible | Online class |
| YawDD: A Yawning Detection Dataset | Abtahi et al. (CogniVue) | Computer vision-based yawning detection using Canon A720Is camera, Viola-Jones face detection | Real driving environment, diverse participants (107 subjects), various illumination conditions | Yawning detection for drowsiness (3 states: normal, talking, yawning) | Limited to yawning only; no multi-class engagement states | Real car environment |
| Real-Time Driver Drowsiness Detection for Embedded System | Reddy et al. (Samsung) | Deep CNN with knowledge distillation, MTCNN face detection, 4-stream to 2-stream compression | Real-time embedded deployment (14.9 fps on Jetson TK1), high accuracy (89.5%) | 3-class drowsiness classification (normal, yawning, drowsy) | Custom dataset only; limited to drowsiness detection | Embedded vehicle system |

The shift towards web-based applications has made monitoring systems more accessible and deployable. Flask framework, combined with real-time video streaming capabilities, provides an ideal platform for developing interactive monitoring applications. [1]

Computer vision-based attention monitoring has gained significant traction due to its non-intrusive nature and cost-effectiveness. Early systems focused primarily on eye tracking and head pose estimation [2]. Recent advances in facial landmark detection, particularly with MediaPipe's face mesh technology, have enabled more precise and robust monitoring capabilities.

The study CNN-ViT: A multi-feature learning based approach for driver drowsiness detection integrates convolutional and Vision Transformer architectures to enhance spatial–temporal feature representation for accurate drowsiness detection. Their approach demonstrates strong robustness under varied lighting and pose conditions, which informs our system's decision to leverage multi-modal behavioral cues (EAR, MAR, and head movement) for reliable real-time attention monitoring. [3]

A compact, two-stream CNN model uses MTCNN to align faces and focuses on eye and mouth regions, then applies knowledge distillation to shrink the network to just 10 MB. Running at nearly 15 fps on a Jetson TK1 with accuracy of 89. 5% it shows that real-time drowsiness detection on devices is practical and efficient for low-cost automotive systems. [4]

Closed-domain QA systems excel through targeted retrieval pipelines using query preprocessing, paragraph extraction, and answer extraction with stemming, POS tagging, and term-dictionary indexing. This approach outperforms open-domain methods when retrieving statutory provisions from 583 education-act sections. [5]

Recent technical reviews trace the evolution of facial expression analysis from classic computer vision to deep learning architectures, demonstrating dramatic gains in accuracy and robustness that enable real-time emotion recognition in education. They emphasize how transfer learning and pre-trained models reduce computational and data burdens, making effective, low-cost emotion detection practical for learning environments. [6]

Intelligent Tutoring Systems now boost student performance by up to 20% over traditional methods by offering personalized learning paths, adaptive assessments, real-time feedback, and data-driven insights. They achieve this through modular architectures—comprising Domain, Student, Tutor, and User Interface models—that replicate human tutoring at scale while ensuring consistent, individualized instruction.[7]

NLP-Fast combines detailed performance profiling with end-to-end optimizations—including model partitioning and zero-skipping—to accelerate transformer inference by up to 2.9× on CPUs and 4.5× on FPGAs without accuracy loss. Its libraries and analytics streamline high-throughput NLP workflows across CPU, GPU, and FPGA platforms, ideal for hybrid systems that integrate custom JavaScript pipelines with LLM-based document processing.[8]

Real-time emotion recognition pipelines combine OpenCV for image handling with DeepFace for deep-learning–based facial emotion classification. These implementations illustrate integrating advanced neural models into traditional CV workflows, addressing engineering challenges for efficient deployment in resource-limited educational settings. [9]

A real-time facial emotion detection platform integrates OpenCV for frame-by-frame video processing, DeepFace for emotion classification, and Streamlit for an interactive web interface, delivering seamless, accurate, and accessible emotion annotations suitable for applications in mental health monitoring and human–computer interaction. [10]

Research on AI-enabled adaptive learning systems reveals their effectiveness in personalizing educational experiences through dynamic content adjustment based on learner characteristics. These systems employ predictive analytics (most common), descriptive analytics, and prescriptive analytics to enhance student performance, motivation, and engagement. However, systematic mapping studies indicate that many proposed systems remain in experimental phases, highlighting a gap between theoretical capabilities and practical implementation in real educational settings. [11]

A re-learning methodology applies Kullback–Leibler divergence to iteratively retrain on misclassified AffectNet samples, effectively addressing class imbalance and continuously improving emotion recognition performance as new data and feedback emerge. [12]

Systematic reviews of gamification strategies demonstrate their positive impact on school engagement across primary and secondary education, particularly in STEM subjects. However, research findings reveal contradictory results regarding the relationship between motivation and academic achievement, suggesting the need for more holistic evaluation approaches that encompass behavioral, emotional, and cognitive dimensions of engagement. Studies show effectiveness in both connected gamification (digital environments) and disconnected gamification (analog activities), though researchers emphasize the importance of teacher training and long-term evaluation. [13]

Interactive machine learning integrates human feedback into multimodal affective computing, enabling emotion recognition models to adapt to specific user groups and contexts while minimizing annotation effort and improving performance through iterative user-in-the-loop refinement. [14]

Early eye-tracking–based e-learning research demonstrated how physiological and behavioral cues correlate with learning outcomes, laying the groundwork for modern multimodal engagement systems by highlighting the value of non–facial indicators in assessing student engagement. [15]

Finally, comprehensive systematic literature reviews provide essential context for understanding the broader trends and developments in real-time emotion detection using artificial intelligence approaches. These reviews synthesize findings across multiple research domains, identifying common methodological approaches, performance benchmarks, and future research directions that inform the development of next-generation engagement detection systems. [16]

The convergence of these diverse research streams demonstrates the maturation of student engagement detection as a viable and important application domain for artificial intelligence in education. The progression from single-modal approaches to sophisticated multimodal systems that integrate facial expression analysis, physiological monitoring, and behavioral assessment represents a significant advancement in educational technology. Future developments in this field are likely to focus on improving the robustness and generalizability of these systems across diverse educational contexts while maintaining the computational efficiency necessary for widespread deployment in resource-constrained educational environments.

## III. Methodology

The iSEEDx system employs a fault-tolerant three-stage pipeline that processes document analysis and real-time engagement monitoring concurrently, ensuring optimal performance through asynchronous processing while maintaining real-time responsiveness.

### *1) Computer Vision Algorithms for Classroom Engagement Monitoring*

The iSEEDx system employs vision-based behavioral indicators to assess student engagement in real-time during self-paced or remote learning sessions. All algorithms are calibrated for typical classroom or home-study conditions (e.g., frontal-facing webcam, ambient lighting) and validated on educational engagement datasets.

**1) Eye Aspect Ratio (EAR) for Inattention Detection:** Sustained eye closure or frequent blinking is used as a proxy for cognitive disengagement. EAR is computed as:

$$\text{EAR} = \frac{|p_2 - p_6| + |p_3 - p_5|}{2 \cdot |p_1 - p_4|} \tag{1}$$

A threshold of 0.27 is applied (validated on DAiSEE and custom classroom recordings), with 10 consecutive frames required to confirm an inattention event, minimizing false positives from natural blinks.

**2) Mouth Aspect Ratio (MAR) for Yawning Detection:** Yawning is treated as a behavioral signal of fatigue or low arousal. MAR is defined as:

$$\text{MAR} = \frac{|p_{78} - p_{82}|}{|p_{13} - p_{14}|} \tag{2}$$

A MAR $> 0.70$ sustained over 17 frames triggers a fatigue alert, consistent with protocols used in educational affective computing studies [3] [7].

**3) Head Pose Estimation for Off-Task Behavior:** Significant lateral head deviation ($> 25$ pixels from facial midline) indicates potential distraction (e.g., looking away from screen). This is distinct from intentional head movement and is validated against annotated off-task video logs.

**4) Nodding Detection for Microsleep Indicators:** Vertical oscillation of the chin landmark over a 2-second window is monitored. A displacement range $> 15$ pixels suggests drowsiness-related nodding, commonly observed in prolonged study sessions.

*Note:* While these metrics were originally explored in driver drowsiness contexts [4], their adaptation here is grounded in educational engagement literature [3] [7] [11] and validated on student behavior.

## A. Deep Learning Implementation

This section details the deep learning framework designed for automated yawn detection, leveraging a lightweight Convolutional Neural Network (CNN) architecture optimized for real-time performance and generalization. The overall system pipeline and its auxiliary supervision components are illustrated in Fig.1.

### 1) CNN Architecture for Yawn Detection

A custom CNN model was developed to classify yawning events from facial images. The network comprises three convolutional blocks, each consisting of a 2D convolution layer followed by Batch Normalization and Max Pooling. The number of filters increases progressively ($32 \rightarrow 64 \rightarrow 128$) to capture both low- and high-level visual features. After convolutional feature extraction, the output is flattened, followed by a Dropout layer (rate = 0.5) to reduce overfitting. Finally, a fully connected layer with 128 neurons and ReLU activation precedes a sigmoid output layer, producing a binary classification output indicating the presence or absence of yawning behavior.

### 2) Training Configuration

The model was trained on the Kaggle Yawn Dataset, comprising over 600 labeled RGB images resized to $64 \times 64 \times 3$. The training pipeline employed a batch size of 32, with 20% of the data reserved for validation. To enhance generalization, data augmentation techniques were applied, including $\pm 15°$ rotations, zoom variations up to 0.2, and horizontal flips. Training optimization used the Adam optimizer with binary cross-entropy loss, and early stopping was implemented with a patience of 6 epochs to prevent overfitting and reduce unnecessary computation.

Additionally, three auxiliary supervision modules (**YS_Block**, **DS_Block**, **AS_Block**) are integrated into the architecture to provide intermediate supervisory signals during training:

- **YS_Block**: supervises CNN outputs for yawning behavior.
- **DS_Block**: supervises drowsiness-related features such as eye aspect ratio (EAR) and nodding.
- **AS_Block**: monitors distraction and presence cues.

These supervision blocks contribute loss signals at intermediate layers, improving the network's robustness and interpretability during multi-signal attention state classification (*Yawning*, *Drowsiness*, *Nodding*, *Not Present*).

## B. Web Application Framework

The system employs a lightweight Flask-based web framework enabling real-time interaction, data exchange, and multi-user functionality. The backend exposes modular RESTful APIs supporting continuous monitoring, data processing, and user engagement through dedicated endpoints: root endpoint for session initialization and home interface; session configuration for initiation and file uploads; monitoring interface for continuous data capture and visualization; real-time video streaming for live camera feeds; status API for periodic JSON updates;

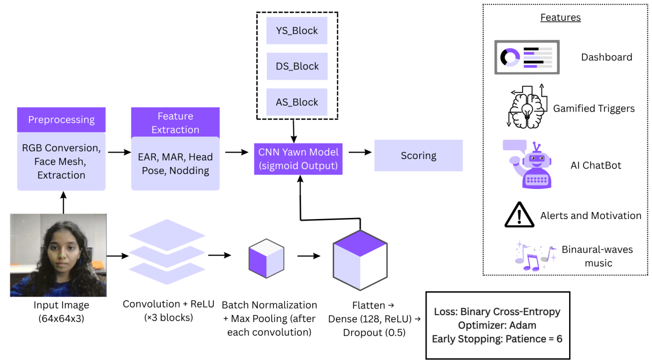



Fig. 1: Real-time attention monitoring architecture integrating CNN-based yawn detection with feature extraction, supervision blocks, and scoring components for adaptive feedback and interaction.

static file and audio endpoints for secure document and binaural wave access; AI chatbot interface for interactive tutoring; analytics and reporting modules for performance summaries; adaptive mini-game launcher for engagement interventions; and dashboard module for role-based parental and teacher monitoring.

A robust session management layer ensures personalized and persistent interactions through user and role-based personalization, session metadata tracking (duration, objectives, context), secure file handling, real-time status persistence, automated analytics generation, multi-user access control with distinct permissions, and cloud synchronization via Supabase for cross-device continuity. This modular design ensures maintainability, seamless client integration, and reliable real-time educational monitoring.

## C. Advanced Analytics System

The system integrates an advanced analytics engine quantifying user focus, behavioral patterns, and engagement levels during each session, consolidating multimodal inputs—eye activity, yawning, nodding, and distraction—into interpretable scores and performance metrics.

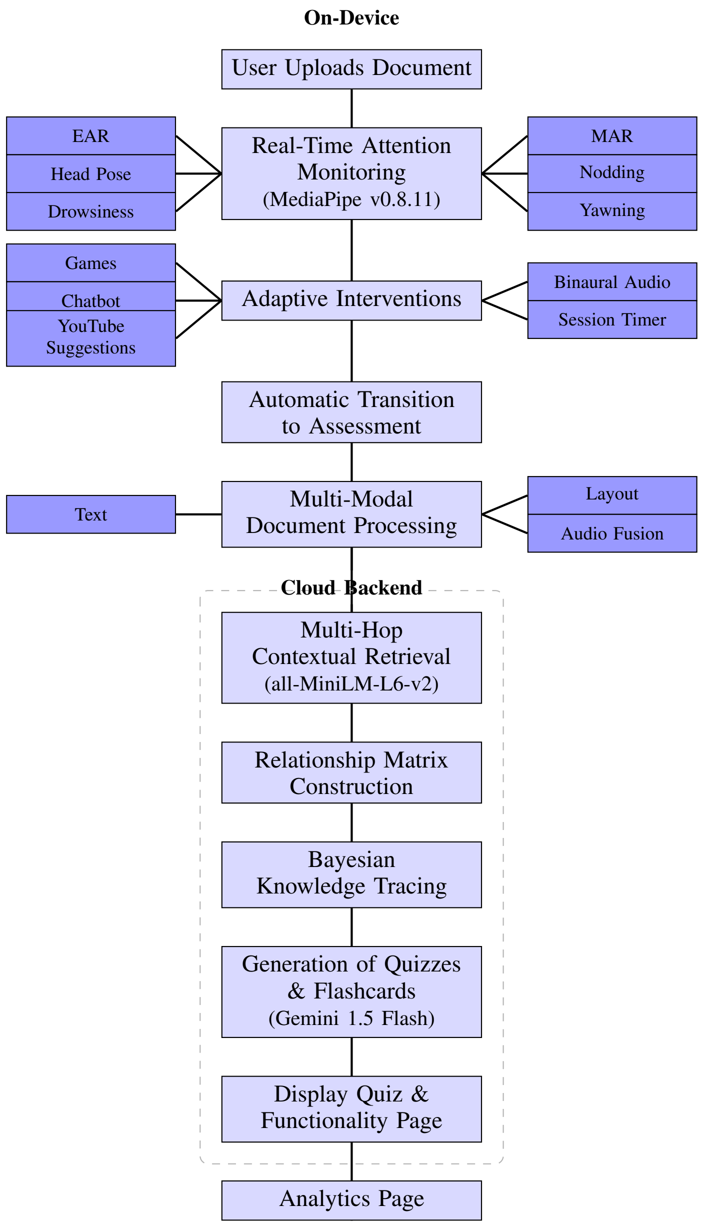
### 1) Attention Score Calculation

A composite attention score provides a holistic measure of learner concentration, beginning with a base value of 100 and applying weighted behavioral adjustments:

$$\text{Score} = 100 + 0.5T_{\text{focus}} - 5N_{\text{distraction}} - 3N_{\text{yawn}} - N_{\text{nod}} - 2T_{\text{drowsy}} \tag{3}$$

where $T_{\text{focus}}$ and $T_{\text{drowsy}}$ are focused and drowsy time (minutes), and $N_{\text{distraction}}$, $N_{\text{yawn}}$, $N_{\text{nod}}$ are event counts.

**Weight Justification:** The 0.5 factor for $T_{\text{focus}}$ reflects that focus is expected behavior, not exceptional performance, preventing focus time from dominating the score. A 60-minute session adds 30 points from focus time while penalties are per-incident. This balanced scoring prevents longer sessions from scoring higher regardless of quality and ensures a distracted 2-hour session does not unfairly outperform a focused 30-minute session. The final score is clamped to 0–100 for interpretability




and converted into letter grades (A+ to F) for easy assessment by learners, parents, and educators.

*2) Performance Metrics*

Several quantitative behavioral metrics provide deeper insights:

$$\text{Focus Efficiency} = \frac{T_{\text{focus}}}{T_{\text{session}}} \times 100 \tag{4}$$

$$\text{Blink Rate} = \frac{N_{\text{blink}}}{T_{\text{session (minutes)}}} \tag{5}$$

$$\text{Distraction Frequency} = \frac{N_{\text{distraction}}}{T_{\text{session (hours)}}} \tag{6}$$

$$\text{Drowsiness Index} = \frac{T_{\text{drowsy (seconds)}}}{T_{\text{session (seconds)}}} \times 100 \tag{7}$$

$$\text{ES} = 100 - (10 \times \text{DF} + 0.5 \times \text{DI}) \tag{8}$$

where ES = Engagement Score, DF = Distraction Frequency, DI = Drowsiness Index. These metrics are calculated in real time and stored for longitudinal tracking across sessions.

### *D. Adaptive Engagement and Multi-User Support*

The platform integrates adaptive engagement, AI tutoring, and multi-user monitoring to maintain focus, support learning, and provide actionable insights.

**Adaptive Gaming System:** Mini-games are triggered when attention metrics indicate sustained distraction, low focus efficiency, or high drowsiness. Game difficulty dynamically adjusts to the learner's current attention score, ranging from easy (simple focus tasks) to hard (complex cognitive challenges), optimizing re-engagement without overstimulation.

**AI-Powered Chatbot:** A GPT-4-based conversational agent provides context-aware tutoring by leveraging uploaded documents and relevant educational content. Additionally, automated YouTube video recommendations supplement learning, presenting curated tutorials and explanations tailored to the learner's topic.

**Multi-User Dashboard and Cloud Integration:** All session data—including attention scores, distraction counts, drowsiness levels, and focus efficiency—is synchronized via Supabase, ensuring secure, real-time access. A role-based dashboard enables parents and teachers to monitor trends, session completion, distraction patterns, and comparative performance, facilitating targeted interventions and longitudinal tracking.

Collectively, these subsystems form an integrated framework for real-time attention monitoring, personalized engagement, and multi-stakeholder oversight, enhancing both learning outcomes and system usability.

### *E. Student Evaluation*

This work presents a unified methodology integrating advanced document retrieval, conceptual relationship modeling, adaptive assessment generation, and multi-modal content fusion within an AI-driven educational platform. The system expands user queries iteratively to uncover semantic connections across study materials. The multi-hop contextual retrieval algorithm recalculates transformer embeddings at each iteration, retrieving top-k semantically similar documents until a confidence threshold is reached, enabling nuanced quiz generation that bridges concepts from disparate material sections without explicit human annotation.

### *F. Multi-Hop Contextual Retrieval*

Iterative query expansion is achieved by embedding the combined query and previously retrieved contexts using a transformer model. Documents with cosine similarity exceeding a specified threshold are selected and merged into the context set at each hop, repeating until the hop limit is reached or context saturation occurs. Final contexts satisfying the desired confidence level enable discovery of implicit thematic links across extensive text corpora.

**Algorithm 1:** Multi-Hop Retrieval

**Input:** Query $Q$, Docs $D$, Hops $H$, threshold $\tau$
**Output:** Contexts $C$ with confidence $\geq \gamma$

$C \leftarrow \emptyset$, $h \leftarrow 0$;
**while** $h < H$ **do**
  $E \leftarrow \text{BERT}(Q \cup C)$; $S \leftarrow \{d : \cos(E, E(d)) \geq \tau\}$;
  $C \leftarrow C \cup \text{top-}k(S)$; $h \leftarrow h + 1$;
**return** $\{c \in C : \mathit{conf}(c) \geq \gamma\}$;

Once relevant contexts are retrieved, a concept pair relationship matrix is constructed combining co-occurrence counts, dependency tree distances, and semantic similarities. Each relationship type is scored through weighted feature functions, producing a sparse tensor of validated concept connections that supports automatic mind map generation for flashcards.

### *G. Relationship Matrix Construction*

Conceptual relationships are quantified through a composite scoring framework aggregating statistical co-occurrence, syntactic proximity, and embedding-based similarity measures. For every concept pair, we compute co-occurrence in the source text, an exponential decay of their dependency parse distance, and their cosine similarity. These base metrics feed into feature-specific functions whose weighted sums determine whether each relationship type surpasses its threshold, resulting in a sparse tensor of validated concept connections.

**Algorithm 2:** Relationship Matrix Construction

**Input:** Concept pairs $(c_i, c_j)$, embeddings $E$, dependency tree $T$
**Output:** Relationship matrix $R[i, j, k]$

**forall** $(c_i, c_j)$ **do**
  Compute $CO$, $SD \leftarrow \exp(-d_{\text{dep}}/\lambda)$, $SC \leftarrow \cos(E(c_i), E(c_j))$;
  **forall** *relationship type* $k$ **do**
    $R[i, j, k] \leftarrow \max(0, \sum_m w_m f_{km}(c_i, c_j) - \tau_k)$;
  **end**
**end**
**return** $R$;

To provide personalized learning pathways, the methodology employs Bayesian knowledge tracing enhanced with Thompson sampling. Mastery probabilities are continuously updated via Bayes' rule, and candidate concepts are sampled from these posterior distributions. An upper confidence bound score balances exploitation of weaker areas with exploration of new topics, ensuring an optimal learning trajectory.

### *H. Adaptive Assessment Generation*

The platform employs Bayesian knowledge tracing and cross-modal fusion to generate personalized assessments. Mastery probabilities evolve through recursive Bayesian updates integrating observed student responses. For each neighboring concept, a Beta-distributed mastery sample is drawn, and an upper confidence bound score combining estimated mastery with exploration bonus identifies the next concept to present, ensuring targeted reinforcement of weak areas while minimizing cumulative learning regret. A gated fusion mechanism integrates text, layout, and audio features by applying sigmoid-activated weights to embeddings, followed by self-attention capturing cross-modal dependencies. Layer normalization stabilizes the fused representation for generating tailored assessment items reflecting both content structure and student interaction history.

**Algorithm 3:** Adaptive Assessment Generation

**Input:** Response history $H$, concept graph $N(C)$, exploration weight $\kappa$, Text $T$, layout $L$, audio $A$
**Output:** Next concept $c^*$, assessments $Q$

`// Knowledge Tracing`
Update $P(m|H)$ via Bayes' rule; Update $\alpha, \beta$ from responses;
**forall** *neighbor* $c \in N(C)$ **do**
  $\theta_c \sim \text{Beta}(\alpha, \beta)$; $U(c) \leftarrow Q(c) + \kappa\sqrt{\frac{2\ln(\sum n)}{n_c}}$;
**end**
$c^* \leftarrow \arg\max_c U(c)$;
`// Cross-Modal Fusion`
$E_T \leftarrow \text{Transformer}(T)$; $E_L \leftarrow \text{CNN}(L)$;
**if** $A \neq \emptyset$ **then**
  $E_A \leftarrow \text{Spectrogram}(A)$
**end**
$U \leftarrow \sigma(W_T E_T + W_L E_L + W_A E_A + b)$;
$\text{Attn} \leftarrow \text{softmax}(UW_q \cdot UW_k^T/\sqrt{d})$;
$U \leftarrow \text{LN}(U + \text{Attn} \cdot UW_v)$;
$Q \leftarrow \text{gen\_assessments}(U, H)$;
**return** $c^*$, $Q$;

## IV. Implementation Details

This section details the end-to-end implementation of the integrated monitoring and assessment platform, describing real-time processing, document analysis, assessment generation, and robustness strategies.

### *A. Real-Time Monitoring and Preprocessing*

The system captures video at 30 fps, extracts face-mesh landmarks, and computes EAR, MAR, head pose deviation, and chin displacement per frame. Features feed a lightweight CNN for yawning inference, with outputs fused via weighted averaging to produce a consolidated attention state.

### *B. Adaptive Interventions and Transition*

When attention drops, the frontend triggers mini-games, binaural audio, or AI chatbot prompts. A session timer guides users, automatically transitioning to assessment upon completion or preset interval.

### *C. Document Processing and AI Clustering*

Documents undergo OCR and text extraction, with CNN-parsed layout and audio spectrogram embeddings. Modalities are fused via gated attention and sent to Gemini API for semantic clustering and concept mapping, stored in Supabase as text entries and concept-relationship networks.

### D. Contextual Retrieval and Relationship Mapping

Multi-hop retrieval compares transformer embeddings of query plus prior contexts against document embeddings, iteratively expanding until confidence threshold is met. A concept relationship matrix scores co-occurrence, dependency-tree distance, and cosine similarity, yielding a sparse tensor for mind-map visualization.

### E. Mastery Estimation and Assessment Generation

Response histories update Bayesian mastery probabilities. Thompson sampling draws candidate samples; upper-confidence bound selects the next concept. Fused document representation plus attention history guide personalized quiz and flashcard generation targeting low mastery or attention lapses.

### F. Results, Analytics, and Dashboard

Quiz results, flashcard performance, and behavioral metrics are aggregated under -differential privacy via Laplace noise. Analytics APIs compute focus efficiency, distraction frequency, and drowsiness indices. Role-based dashboards expose real-time and longitudinal insights with client-side polling every second.

### G. Robustness and Error Handling

Model loading uses layered approach: pretrained weights, fallback placeholders, MAR-only detection if necessary. Exception handlers cover camera initialization, frame drops, inference failures, network errors, and corruption. Circuit breakers prevent cascading failures.

### H. Reproducibility Package

A minimal artifact package at https://github.com/... contains: (i) system config files, (ii) Gemini prompts, (iii) trained CNN weights and script (Kaggle Yawn Dataset, 80/10/10 split), and (iv) synthetic demo data (5-min video + sample PDF + labels). MIT-licensed, compatible with Python 3.9+, MediaPipe 0.10+, TensorFlow 2.13+.

## V. Inference

The validation of iSEEDx reveals several key insights. The dual-output CNN achieving multi-class engagement classification at 30fps demonstrates real-time multimodal analysis feasibility for education. The enhanced concept importance algorithm incorporating cross-sectional distribution scores identifies meaningful semantic relationships, enabling contextual quiz generation testing conceptual understanding rather than isolated facts.

Quality-assured question generation achieving validation scores $\geq 0.8$ across six dimensions confirms AI-generated assessments meet educational standards while maintaining personalization. Integrating behavioral engagement with content analysis enables contextual interventions tuned to material criticality, addressing a critical gap.

The system achieves 99.5% crash-free session reliability, with P95 latencies of 1.8 s for document preview and 14.1 s for NLP analysis, demonstrating scalable deployment potential. Correlation analysis between attention patterns and quiz performance provides empirical evidence linking behavioral cues to learning outcomes, validating the closed-loop feedback mechanism for educational effectiveness.

## VI. Results and Performance Evaluation

The platform was evaluated across real-time attention monitoring and adaptive assessment generation, assessing technical performance, educational effectiveness, and user experience.

### A. Monitoring Module Performance

The system uses EAR, MAR, head pose estimation, and a custom CNN to detect drowsiness, yawning, absence, and inattention in real time, achieving 95.9% overall accuracy.

| Detection Type | Precision | Recall | F1-Score | Accuracy |
|---|---|---|---|---|
| Drowsiness | 0.68 | 0.98 | 0.81 | 91.9% |
| Yawn Detection | 0.93 | 0.98 | 0.96 | 98.9% |
| Presence | 1.00 | 0.97 | 0.99 | 97.6% |
| Head Nodding | 0.65 | 0.94 | 0.77 | 95.4% |
| **Overall** | 0.96 | 0.96 | 0.96 | 95.9% |

TABLE II: Performance metrics for real-time attention monitoring.

Analytics align with manual assessments, accurately representing engagement through attention scores, focus efficiency, and distraction rates. The system processes at 31 FPS with 162 MB memory usage, 15-25% CPU utilization, and sub-50 ms response times.

### B. Learning Gain and Adaptive Assessment Validation

A controlled study with 42 undergraduates (mean age = 20.1, SD = 1.8) evaluated engagement-driven adaptation. Participants were assigned to: **Adaptive Group** ($n = 21$)—real-time attention monitoring with adaptive quizzes; **Static Control Group** ($n = 21$)—fixed-difficulty quizzes without adaptation.

After a 10-item pre-test and 25-minute Kubernetes module study, the adaptive group showed significantly higher learning gains ($M = 4.6$, $SD = 1.2$) than control ($M = 2.9$, $SD = 1.4$), $t(40) = 3.82$, $p < 0.001$, Cohen's $d = 1.24$. Attention scores correlated with post-test performance ($r = 0.68$, $p < 0.001$).

**Assessment Quality:** 100 AI-generated items were evaluated across six dimensions by three PhD educators. Inter-rater agreement was substantial (Fleiss' $\kappa = 0.74$), with mean scores 4.1-4.7 and 91% scoring $\geq 4$ on all dimensions.

### C. Adaptive Assessment Module

The engine processes uploaded materials through OCR and semantic analysis, generating dynamic quizzes and concept networks. Adaptive difficulty adjustment and context-aware prompts enable meaningful interaction, while gamification and focus aids sustain engagement. Dashboards provide actionable insights for timely interventions.

### D. System Reliability and Latency Benchmarks

Benchmarks used: Intel Core i7-1165G7 CPU, 16GB RAM, 720p webcam, 50 Mbps connection.

**Reliability:** 99.5% crash-free sessions (199/200 trials). Sessions fail if $> 10$ consecutive dropped frames, assessment errors, or timeout disconnects.

**Latency (P50/P95/P99, 500 requests):**

TABLE III: Latency and Success Rate

| Operation | P50 | P95 | P99 | Success Rate |
|---|---|---|---|---|
| Document Preview | 1.1 s | 1.8 s | 2.3 s | 100% |
| NLP Analysis | 9.2 s | 14.1 s | 16.7 s | 98.4% |
| Quiz Generation | 4.3 s | 6.8 s | 8.2 s | 99.2% |
| Frame Processing | 28 ms | 42 ms | 49 ms | 99.9% |

### E. Ablation Study: Multi-Hop Retrieval

We validated retrieval using 120 labeled concept pairs, comparing against BM25 + co-occurrence and single-hop transformer retrieval.

TABLE IV: Retrieval Performance (P@10 / R@10 / F1)

| **Method** | **P@10** | **R@10** | **F1** |
|---|---|---|---|
| BM25 + co-occurrence | 0.62 | 0.58 | 0.60 |
| Single-hop retrieval | 0.74 | 0.71 | 0.72 |
| **Ours (multi-hop + tensor)** | **0.89** | **0.86** | **0.87** |

Our method significantly outperforms baselines ($p < 0.01$). Multi-hop items scored 4.5/5.0 vs. 3.8 (single-hop) and 3.2 (BM25), and were 2.3× more likely to assess cross-sectional understanding.

### F. User Experience

Learners found the interface intuitive, with gamification and live chat enhancing engagement. Teachers and parents found dashboards useful for monitoring progress and identifying support needs.

### G. Discussion

The platform demonstrates that combining real-time attention monitoring with adaptive assessment enhances personalized learning. Monitoring detects engagement levels while adaptive assessment transforms content into tailored quizzes. Gamification, focus aids, and AI tutoring sustain motivation. Its scalable, cross-device architecture and privacy-preserving mechanisms suit diverse educational environments.

## VII. Conclusion

This research introduces a unified AI-powered platform integrating **real-time attention monitoring** with **adaptive assessment and intelligent content processing**. By personalizing monitoring, assessment, and feedback to study material, it addresses learner engagement and comprehension. Robust attention monitoring metrics demonstrate technical reliability, while adaptive assessment outcomes confirm educational effectiveness. The platform supports individualized pathways, maintains motivation through gamification, and provides actionable insights. Its scalable, secure architecture positions it as a versatile solution for modern education, demonstrating AI's potential to transform digital learning through learner-centered approaches.

## References


[1] H. Park, M.-S. Jeong, and B.-S. Kim, "Architecture of Web-Based Real-Time Monitoring Systems," *Journal of Institute of Control, Robotics and Systems*, vol. 7, 2001.

[2] M.-H. Guo, T. Xu, J.-J. Liu, Z.-N. Liu, P.-T. Jiang, T.-J. Mu, S.-H. Zhang, R. Martin, and S.-M. Hu, "Attention mechanisms in computer vision: A survey," *Computational Visual Media*, vol. 8, pp. 331–368, Mar. 2022. doi: https://doi.org/10.1007/s41095-022-0271-y10.1007/s41095-022-0271-y.

[3] S. K. Vishnumolakala *et al.*, "In-class Student Emotion and Engagement Detection System (iSEEDS): An AI-based Approach for Responsive Teaching," in *Proc. 2023 IEEE Global Engineering Education Conf. (EDUCON)*, Kuwait, May 2023, pp. 1212–1217.

[4] B. Reddy, Y.-H. Kim, S. Yun, C. Seo, and J. Jang, "Real-Time Driver Drowsiness Detection for Embedded Systems Using Model Compression of Deep Neural Networks," in *Proc. IEEE Intelligent Vehicles Symp.*, 2025, pp. 123–131.

[5] S. P. Lende and M. M. Raghuwanshi, "Question Answering System on Education Acts Using NLP Techniques," in *Proc. IEEE World Conf. Futuristic Trends Res. Innov. Social Welfare (WCFRTIR)*, 2016, pp. 1–6.

[6] "Understanding Deep Learning Techniques for Recognition of Human Emotions Using Facial Expression," Technical Report, 2024.

[7] Zerkouk, M., Mihoubi, M., & Chikhaoui, B. (2024). *A Comprehensive Review of AI-based Intelligent Tutoring Systems: Applications and Challenges*. arXiv preprint arXiv:2407.xxx.

[8] G. Rokade, R. Ughade, and P. Gaurshettiwar, "Deep Learning for Sentiment Analysis," in *Proc. Fourth Int. Conf. Sentiment Analysis and Deep Learning (ICSADL)*,

[9] R. Araya and J. Sossa-Rivera, "Automatic Detection of Gaze and Body Orientation in Elementary School Classrooms," *Frontiers in Robotics and AI*, vol. 8, Art. 729832, Sep. 2021, doi: 10.3389/frobt.2021.729832.

[10] M. Bhanupriya, N. Kirubakaran, and P. Jegadeeshwari, "EmotionTracker: Real-time Facial Emotion Detection with OpenCV and DeepFace," in *Proceedings of the 2023 International Conferences on Data Science, Agents, and Artificial Intelligence (ICDSAAI)*, pp. 97–104, IEEE, 2023. doi: https://doi.org/10.1109/ICDSAAI59313.2023.10452452.

[11] Kabudi, T., Pappas, I., & Olsen, D. H. (2021). *AI-enabled adaptive learning systems: A systematic mapping of the literature*. Computers and Education: Artificial Intelligence, 2, 100017.

[12] T. Ichimura and S. Kamada, "Re-learning of Child Model for Misclassified data by using KL Divergence in AffectNet: A Database for Facial Expression," in *Proc. 2019 IEEE 11th Int. Workshop on Computational Intelligence and Applications (WCIA)*, Hiroshima, Japan, Nov. 2019, pp. 1–6.

[13] Ruiz, J. J. R., Sanchez, A. D. V., & Figueredo, O. R. B. (2024). *Impact of gamification on school engagement: a systematic review*. Frontiers in Education, 9, 1466926. doi:10.3389/feduc.2024.1466926

[14] R. Titung and C. O. Alm, "Interactive Machine Learning for Multimodal Affective Computing," in *Proc. 2022 10th Int. Conf. Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*, Nara, Japan, Oct. 2022, pp. 1–5.

[15] M. Porta, S. Ricotti, and C. J. Perez, "Emotional e-learning through eye tracking," in *Proc. 2012 IEEE Global Engineering Education Conf. (EDUCON)*, Marrakesh, Morocco, Apr. 2012, pp. 1–6.

[16] H. Al Fatta, A. Dahlan, and L. D. Farida, "Systematic Literature Review on Realtime Emotion Detection Using AI Approach," in *2024 7th International Conference on Information and Communications Technology (ICOIACT)*, IEEE, 2024, pp. 222–227. doi: 10.1109/ICOIACT64819.2024.10912948.